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PID: A19136116

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Find_A_Gene_Project

Q1.

- Gene: **KIF11 kinesin-like protein**
- Accession: NP_004514.2
- Organism: *Homo Sapiens*
- Query length: 1056 amino acids
- BLAST type: tblastn (protein → nucleotide)

I selected the human kinesin-like protein KIF11 (NP_004514.2) as my query protein. KIF11 is a motor protein involved in mitotic spindle formation and chromosome separation during cell division. This protein plays an essential role in spindle assembly and proper progression through mitosis. The amino acid sequence of human KIF11 was used as the query sequence for subsequent BLAST searches against translated nucleotide databases in order to identify homologous but potentially unannotated sequences from poorly characterized organisms.

Q2.

Output/Screenshot

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☑	TSA: Podocoryna carnea Podocoryna_60013.0_Transcript_2/0_Confidence_56_Length_4346 transcribed RNA sequence	1133	1133	100%	0.0	54.27%	4346	GCHV01026134.1
☑	TSA: Podocoryna carnea comp127897_c0_seq3 transcribed RNA sequence	1133	1133	100%	0.0	54.27%	3968	GBEH01003494.1
☑	TSA: Podocoryna carnea comp127897_c0_seq8 transcribed RNA sequence	1004	1004	91%	0.0	52.86%	3572	GBEH01003501.1
☑	TSA: Podocoryna carnea comp127228_c0_seq1 transcribed RNA sequence	724	724	38%	0.0	84.20%	1336	GBEH01119169.1
☑	TSA: Zoanthus sp. QL-2018 CL16872.Contig2 transcribed RNA sequence	682	682	89%	0.0	42.83%	3550	GGTW01041159.1
☑	TSA: Zoanthus sp. QL-2018 CL16872.Contig1 transcribed RNA sequence	682	682	89%	0.0	42.83%	3550	GGTW01041158.1
☑	TSA: Aurelia aurita comp196491_c0_seq1 transcribed RNA sequence	681	681	89%	0.0	41.31%	3447	GBRG01140943.1
☑	TSA: Rhopilema esculentum c73317_g1_i2 transcribed RNA sequence	681	681	99%	0.0	39.60%	3673	GEMS01083823.1
☑	Rhopilema esculentum c129057.graph_c1	681	681	99%	0.0	39.51%	3407	GGZZ01027795.1
☑	Pocillopora damicornis breed cauliflower coral isolate corallarvae Unigene0001813	680	680	53%	0.0	60.10%	5143	GJYV01001030.1
☑	TSA: Stylophora pistillata Locus_14008_Transcript_1/1_Confidence_0.000_Length_3860 transcribed RNA sequence	677	677	53%	0.0	59.93%	3860	GARY01013264.1
☑	Pocillopora acuta TRINITY_DN2475_c0_g1_i9	674	731	58%	0.0	59.58%	4773	GJER01007804.1
☑	Pocillopora acuta TRINITY_DN2475_c0_g1_i6	671	730	58%	0.0	59.48%	4414	GJER01007808.1

TSA: Podocoryna carnea comp127228_c0_seq1 transcribed RNA sequence

Sequence ID: GBEH01119169.1 **Length:** 1336 **Number of Matches:** 1

Range 1: 1 to 1212 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
724 bits(1870)	0.0	Compositional matrix adjust.	351/405(87%)	388/405(95%)	1/405(0%)	-2

```
Query  1     MASQPNSSAKKKEEKGKNIQVVVRCRPFNLAERKASAHSIVECDPVRKEVSVRTGGLADK  60
             M+SQ N S  KK++KGKNIQVVVRCRPFN AERKAS++S+++CD  RKE+ VRTGG+ DK
Sbjct  1212  MSSQ-NISVGKKDDKGKNIQVVVRCRPFNNAERKASSYSVMDCDSGRKEIVVRTGGVGDK  1036

Query  61    SSRKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQTGTGKTFTMEGERS  120
             +++KTYTFDMVFG S KQI+VYRSVV PILDEVIMGYNCT+FAYGQTGTGKTFTMEGERS
Sbjct  1035  TAKKTYTFDMVFGPSAKQIEVYRSVVVPILDEVIMGYNCTVFAYGQTGTGKTFTMEGERS  856

Query  121   PNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVSE  180
             PNEE+TWEEDPLAGIIPRTLHQIFEKLT NGTEFSVKVSLLEIYNEELFDLL+PS+DV+E
Sbjct  855   PNEEFTWEEDPLAGIIPRTLHQIFEKLTSNGTEFSVKVSLLEIYNEELFDLLSPSTDVNE  676

Query  181   RLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVFS  240
             RLQMFDDPRNKRGV+IKGLEEI+VHNK EVYQILE+GAAKR TA+TLMNAYSSRSHSVF+
Sbjct  675   RLQMFDDPRNKRGVLIKGLEEISVHNKYEVYQILERGAAKRKTASTLMNAYSSRSHSVFA  496

Query  241   VTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVIT  300
             +TIHM+ETT+DGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVIT
Sbjct  495   ITIHMRETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVIT  316

Query  301   ALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATISPAslnleetlstleYAHRAKNIL  360
             ALVERTPH+PYRESKLTRILQDSLGGRT+TSIIAT+SPAS+NLEETLSTL+YA+RAKNI+
Sbjct  315   ALVERTPHIPYRESKLTRILQDSLGGRTKTSIIATVSPASVNLEETLSTLDYANRAKNIM  136

Query  361   NKPEVNQKLTKKALIKEYTEEIERLKRDLAAAREKNGVYISEENF  405
             NKPEVNQKLTK+ LIKEYTEEIERLKRDL+AAREKNGVYIS EN+
Sbjct  135   NKPEVNQKLTKRTLIKEYTEEIERLKRDLSAAREKNGVYISSENY  1
```

Using the NCBI Transcriptome Shotgun Assembly (TSA) database and the human KIF11 protein sequence (NP_004514.2), I performed a tblastn search restricted to poorly annotated cnidarian organisms. The search identified several significant transcript matches from marine invertebrates and coral-related species.

The selected alignment showed strong homology to the human KIF11 query protein while remaining only partially conserved.

Alignment statistics

- **Score:** 724 bits
- **E-value:** 0.0
- **Query coverage:** 38%
- **Percent identity:** 84.20%

The transcript sequence lacked clear functional annotation and was identified only as a transcribed RNA sequence, making it a suitable candidate for investigation as a potentially novel homologous gene.

Q3.

The selected protein sequence was obtained by translating the nucleotide transcript sequence from Q2 using EMBOSS Transeq. All six reading frames were translated, and the longest open reading frame with minimal stop codons was selected as the predicted protein sequence.

Predicted protein FASTA sequence:

```
>1-1212_1 TSA: Podocoryna carnea comp127228_c0_seq1 transcribed RNA
sequence
```

VVFRRNIHSVFLSSC*KVTFKAFNFLSVFLNECPFGEFLVYLRLVHNVLCPVSIIQCAQG

FLQIDRGRGNSGDDRRLRATSKRVLKDTSELGFSVWYVRGPLHKGCDHSAQS*E*LVDIT

SFSCPFVHSSRATNVLTAGKINQVQFANFDKFFSINSGLPHVNCNCKD*MRAAGVSIH*G

*CSLPLGCTTLQNLINLILVVYRNLFQSFDKNPPLITRIIKHLQPLVNVSRRAQQVKKLL

VVNL*QGHFDREFRSIASQLLKDLMQSSRNDTSQRVLFPCKFFIGRPFTFHCEGFSSSSL

PICKHCTVVSHDNFIQNRNHHTAVHFNLFCRRPKYHIKGVSFLCRLITNSSCAYNNFFSP

RVTVHNTIRRCFAFCIIERSASYNDLNILPLVVLFTNRYILRRH

The predicted protein sequence originates from *Podocoryna carnea* and appears to represent a KIF11-like motor protein homolog identified from an unannotated transcriptome assembly sequence.

Q4.

A BLASTP search was performed using the translated protein sequence from Q3 as the query against the NCBI non-redundant (nr) protein database.

The top matches identified were hypothetical proteins with relatively low sequence identity and limited query coverage. Importantly, no proteins with 100% amino acid identity were identified in the same species.

One representative top hit was:

hypothetical protein EYF80_016160 [Liparis tanakae]

Top alignment statistics

- E-value: 3e-07
- Percent identity: 49.21%
- Query coverage: 16%

Because no identical protein match was identified in the database, and only partial hypothetical protein matches were recovered, this result suggests that the translated sequence from *Podocoryna carnea* likely represents a potentially novel KIF11-like homolog.

select all 2 clusters selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

	Cluster Composition Click the ⊞ to see the cluster contents	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☑	⊞1 member(s), 1 organism(s)	Tanaka's snailfish	hypothetical protein EYF80_016160 [Liparis tanakae]	64.3	64.3	16%	3e-07	49.21%	339	TNN73565.1
☑	⊞1 member(s), 1 organism(s)	softshell	hypothetical protein MAR_015921 [Mya arenaria]	55.1	55.1	11%	8e-05	55.81%	173	WAR21947.1

hypothetical protein EYF80_016160 [Liparis tanakae]

Sequence ID: TNN73565.1 Length: 339 Number of Matches: 1

Range 1: 145 to 207 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
64.3 bits(155)	3e-07	Compositional matrix adjust.	31/63(49%)	45/63(71%)	0/63(0%)

```
Query  215  LITRIIKHLQPLVNVSRRAQQVKKLLVVNL*QGHFDREFRSIASQLLKDLMQSSRNDTSQ  274
            L+ R+++ LQPL +V    A+QV++LLVV+L Q  FDRE  ++   +LLKDL+Q S +D  Q
Sbjct  145  LVARVVEELQPLGDVVAGAEQVEQLLVVDLQQRDFDRELGAVLRKLLKDLVQRSGDDAGQ  204

Query  275  RVL  277
             VL
Sbjct  205  GVL  207
```